THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** 33	$33 \times 33 = 1089$
2	**(c)** 225	$15^2 = 225$
3	**(d)** 289	$17 \times 17 = 289$
4	**(a)** 3^{10}	Add the exponents: $4 + 6 = 10$
5	**(b)** 9	Reciprocal squared
6	**(a)** 1000	$2^{10} = 1024$
7	**(c)** 94	Add the values, subtracting a smaller symbol that comes before a larger one
8	**(c)** subtracted	IV = 4
9	**(d)** 90°	Angle sum of a quadrilateral is 360°
10	**(c)** rhombus	Square has all sides equal
11	**(d)** 100	Ends in two zeros
12	**(a)** 6174	Repeated rearranging and subtracting

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Yes, $1225 = 35 \times 35$, so $\sqrt{1225} = 35$.	1+1
14	-1 for an odd power (1); 1 for an even power (1).	1+1
15	Ten symbols give every number using place value and zero, making arithmetic simple (2).	2
16	A square has four right angles and opposite sides equal (2).	2
17	A number is divisible by 3 if the sum of its digits is divisible by 3 (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Factorise 2025 into prime pairs (2); the square root is 45 (1).	2+1
19	3888 = MMMDCCCLXXXVIII (2); grouped by thousands, hundreds, tens and ones (1).	2+1
20	Adjacent angles are supplementary: $A + B = 180°$, $B + C = 180°$ (2); so $A = C$ (1).	2+1
21	Prime factorise each (1); take the highest powers (1); LCM = 90 (1).	1+1+1
22	Half-diagonals 8, 15 form a right triangle (1); side = 17 cm (1); perimeter = 68 cm (1).	1+1+1
	OR: $5x = 180$, $x = 36$ (1); $A = 118°$, $B = 62°$ (1); $C = 118°$, $D = 62°$ (1).	1+1+1
23	Digit sum condition (2); digits: 0, 3, 6, 9 (1).	2+1
	OR: LCM 360 (1.5); HCF 12 (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) $\left(\frac{2}{3}\right)^{-1} = \frac{3}{2}$ (2). (ii) $5^6 \times 5^2 / 5^{10} = 5^{-2} = \frac{1}{25}$ (3).	2+3
	OR: (i) $9.46 \times 10^{12} \times 10^5 = 9.46 \times 10^{17}$ km (3). (ii) Already in standard form (2).	3+2
25	(i) LCM = 360 s (2). (ii) 9:06 a.m. (1). (iii) 11 times (2).	2+1+2
	OR: (i) Last digit; last two digits; last digit 0/5; last three digits (3). (ii) Appropriate checks (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** 15	$15^3 = 3375$
2	**(d)** 14	$14 \times 14 = 196$
3	**(c)** 4.12×10^{-2}	One non-zero digit before the decimal point
4	**(b)** 10^{15}	Multiply the exponents
5	**(d)** 10^9	A thousand million
6	**(b)** 64	Add digit × place value in base 4
7	**(d)** XCIX	99 = XCIX
8	**(b)** 60	Sexagesimal
9	**(c)** 100°	Angle sum of a quadrilateral is 360°
10	**(a)** bisect each other	Parallelogram property
11	**(a)** 2	Digit sum must be a multiple of 3
12	**(b)** Yes	Digit sum 18 is divisible by 9

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	The sum of the first n odd numbers is n^2 (1); so the sum is $9^2 = 81$ (1).	1+1
14	$3^8 \div 3^6 = 3^2 = 9$ (2).	2
15	Egyptian, Mesopotamian (also Roman, Maya) (1+1).	1+1
16	They bisect each other at right angles (1); they bisect the vertex angles (1).	1+1
17	The difference between the sums of digits at odd and even places is 0 or a multiple of 11 (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$5832 = 2^3 \times 3^6$ (2); cube root $= 18$ (1).	2+1
19	Add in the numerator: 3^{10} (1); subtract: 3^6 (1); value 729 (1).	1+1+1
20	Half-diagonals 9, 12 form a right triangle (1); side $= 15$ cm (1); perimeter $= 60$ cm (1).	1+1+1
21	Prime factorise each (1); take the highest powers (1); LCM $= 240$ (1).	1+1+1
22	5432−2345=3087 (1); 8730−0378=8352 (1); 8532−2358=6174 (1).	1+1+1
	OR: $777 = 3 \times 259 = 3 \times 7 \times 37$ (3).	3
23	For odd m, triplet $(m, \frac{m^2-1}{2}, \frac{m^2+1}{2})$ (1); $m = 9$ gives 40 and 41 (1); $9^2 + 40^2 = 41^2$ (1).	1+1+1
	OR: $180 = 2^2 \times 3^2 \times 5$ (1); multiply by 5 (1); 900, $\sqrt{900} = 30$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) $\frac{21}{28} = \frac{3}{4}$ (2.5). (ii) $\frac{7}{12}$ (2.5).	2½+2½
	OR: (i) $(5, 12, 13)$, $(7, 24, 25)$, $(9, 40, 41)$ (3). (ii) $5^2 + 12^2 = 25 + 144 = 169 = 13^2$ (2).	3+2
25	(i) 58 (2). (ii) 111010_2 (2). (iii) $10001_2 = 17$ (1).	2+2+1
	OR: (i) $2^8 = 256$ (2). (ii) 255 (1). (iii) 11001000_2 (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** 7	Squares end only in 0, 1, 4, 5, 6 or 9
2	**(b)** 400	Sum of first n odd numbers $= n^2 = 400$
3	**(a)** 2^8	Add the exponents: $5 + 3 = 8$
4	**(b)** 7.2×10^4	One non-zero digit before the decimal point
5	**(b)** India	Hindu numerals
6	**(c)** 2024	Add the values, subtracting a smaller symbol that comes before a larger one
7	**(d)** 122	Repeated division of 17 by 3
8	**(a)** 90°	Angle sum of a quadrilateral is 360°
9	**(b)** rhombus	Rhombus property
10	**(b)** 17 cm	Diagonals bisect at right angles; use Pythagoras
11	**(c)** Yes	Digit sum 15 is divisible by 3
12	**(c)** 36	Smallest common multiple

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	A number that is the square of a natural number (1); e.g. 16 and 49 (1).	1+1
14	$\frac{1}{5^3} = \frac{1}{125}$ (2).	2
15	No zero or place value, long strings of symbols (2).	2
16	They are equal (1); they bisect each other (1).	1+1
17	Numbers whose HCF is 1 (1); 8 and 15 (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$64 = 8^2$ (1); so the sum of the first 8 odd numbers (1): $1 + 3 + 5 + 7 + 9 + 11 + 13 + 15$ (1).	1+1+1
19	Add in the numerator: 5^8 (1); subtract: 5^5 (1); value 3125 (1).	1+1+1
20	2499 = MMCDXCIX (2); grouped by thousands, hundreds, tens and ones (1).	2+1
21	Prime factorise each (1); take the highest powers (1); LCM $= 300$ (1).	1+1+1
22	Factorise 4624 into prime pairs (2); the square root is 68 (1).	2+1
	OR: $17576 = 2^3 \times 13^3$ (2); cube root $= 26$ (1).	2+1
23	LHS $= \frac{1}{10}$; RHS $= \frac{1}{2} \times \frac{1}{5} = \frac{1}{10}$ (3).	3
	OR: $2^6 \times 2^{-4} = 2^2 = 4$ (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) a^{m+n}, a^{m-n}, a^{mn} (3). (ii) $= 2^7 \times 5^4/(2^3 \times 5^3) = 2^4 \times 5 = 80$ (2).	3+2
	OR: (i) $2 \times 10^9 \times 3^3 = 5.4 \times 10^{10}$ (2). (ii) 5.4×10^{10} (1). (iii) after 4 hours: 1.62×10^{11} (2).	2+1+2
25	(i) $\sqrt{150^2 + 80^2} = 170$ cm (2). (ii) 460 cm (1). (iii) 340 cm (2).	2+1+2
	OR: (i) Rhombus: equal sides, unequal diagonals meeting at right angles; rectangle: equal diagonals, right angles (3). (ii) Square (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.